Practical Workbook

CT-261

DATABASE MANAGEMENT SYSTEMS

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Batch: **2024**

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Practical Workbook

Database Management Systems

(CT – 261)

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Q1. Create the DEPT table based on the following table instance chart. Confirm that the table is created.

ID NAME Column Name Key Type Nulls/Unique FK Table FK Column

NUMBER VARCHAR2 **Data type** 7 25 **Length.**

SOLUTION:

```
-- Q1

CREATE TABLE DEPT(
ID NUMBER(7) PRIMARY KEY,
NAME VARCHAR2(25)
);
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table created.
```

0.03 seconds

```
DESCRIBE DEPT
```

Results | Explain | Describe | Saved SQL | History

Object Type **TABLE** Object **DEPT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPT	ID	NUMBER	-	7	0	1	-	-	-
	NAME	VARCHAR2	25	-	-	-	✓	-	-

Q2. Populate the DEPT table with data from the DEPARTMENTS table. Include only columns that you need.

CREATING DEPARTMENTS TABLE (As it does not exist):

```
CREATE TABLE DEPARTMENTS (
    DEPARTMENT_ID NUMBER(4) PRIMARY KEY,
    DEPARTMENT_NAME VARCHAR2(30) NOT NULL,
    MANAGER_ID NUMBER(6),
    LOCATION_ID NUMBER(4)
);
```

INSERTING VALUES IN THE DEPARTMENTS TABLE:

```
INSERT INTO DEPARTMENTS (DEPARTMENT_ID, DEPARTMENT_NAME) VALUES (10, 'ADMINISTRATION');
INSERT INTO DEPARTMENTS (DEPARTMENT_ID, DEPARTMENT_NAME) VALUES (20, 'MARKETING');
INSERT INTO DEPARTMENTS (DEPARTMENT_ID, DEPARTMENT_NAME) VALUES (30, 'PURCHASING');
INSERT INTO DEPARTMENTS (DEPARTMENT_ID, DEPARTMENT_NAME) VALUES (40, 'HUMAN RESOURCES');
```

SOLUTION:

```
-- Q2

INSERT INTO DEPT (ID, NAME)
SELECT DEPARTMENT_ID,DEPARTMENT_NAME
FROM DEPARTMENTS;
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
4 row(s) inserted.

0.20 seconds
```

```
SELECT * FROM DEPT;
```

Results | Explain | Describe | Saved SQL | History

ID	NAME
40	HUMAN RESOURCES
10	ADMINISTRATION
20	MARKETING
30	PURCHASING

4 rows returned in 0.14 seconds Download

Q3. Create the EMP table based on the following table instance chart. Confirm that the table is created. ID LAST_NAME FIRST_NAME DEPT_ID

Column Name

Key

Type

Nulls/Unique

FK Table

FK Column

NUMBER VARCHAR2 VARCHAR2 NUMBER

Data type Length 7 25 25 7

SOLUTION:

```
-- Q3

CREATE TABLE EMP(
ID NUMBER (7) PRIMARY KEY,
LAST_NAME VARCHAR2(25),
FIRST_NAME VARCHAR2(25),
SALARY NUMBER(9,2),
DEPT_ID NUMBER(7)
);
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table created.

0.03 seconds
```

```
DESCRIBE EMP
```

Results | Explain | Describe | Saved SQL | History

EMP	ID	NUMBER	-	7	0	1	-	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	FIRST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	9	2	-	✓	-	-
	DEPT_ID	NUMBER	-	7	0	-	✓	-	-

Q4. Modify the EMP table to allow for longer employee last names. Confirm your modification.

SOLUTION:

```
-- Q4

ALTER TABLE EMP
MODIFY (LAST_NAME VARCHAR2(50));
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table altered.
```

0.02 seconds

Q5. Confirm that both the DEPT and EMP tables are stored in the data dictionary.

(**Hint:** USER_TABLES)

SOLUTION:

```
-- Q5

SELECT TABLE_NAME
FROM USER_TABLES
WHERE TABLE_NAME IN ('DEPT','EMP');
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

TABLE_NAME
DEPT
EMP

2 rows returned in 0.06 seconds Download

Q6. Create the EMPLOYEES2 table based on the structure of the EMPLOYEES table. Include only the EMPLOYEE_ID , FIRST_NAME , LAST_NAME , SALARY , and DEPARTMENT_ID columns. Name the columns in your new table ID , FIRST_NAME , LAST_NAME , SALARY , and DEPT_ID , respectively.

SOLUTION:

```sql
-- Q6

CREATE TABLE EMPLOYEES2 AS
SELECT
ID,
FIRST_NAME,
LAST_NAME,
SALARY,
DEPT_ID
FROM EMP;
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table created.
```

0.05 seconds

```sql
DESCRIBE EMPLOYEES2
```

Results | Explain | Describe | Saved SQL | History

Object Type TABLE Object EMPLOYEES2

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEES2	ID	NUMBER	-	7	0	-	✓	-	-
	FIRST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	9	2	-	✓	-	-
	DEPT_ID	NUMBER	-	7	0	-	✓	-	-

Q7. Drop the EMP table.

SOLUTION:

```
-- Q7

DROP TABLE EMP;
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table dropped.
```

0.97 seconds

Q8. Rename the EMPLOYEES2 table as EMP.

SOLUTION:

```
-- Q8

RENAME EMPLOYEES2 TO EMP;
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Statement processed.
```

0.04 seconds

Q9. Add a comment to the DEPT and EMP table definitions describing the ables. Confirm your additions in the data dictionary.

SOLUTION:

```sql
-- Q9

COMMENT ON TABLE DEPT IS 'This table contains information of the department';
COMMENT ON TABLE EMP IS 'This table contains information of all the employees';
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Statement processed.

0.01 seconds
```

CONFIRMING THE ADDITIONS IN DATA DICTIONARY:

```sql
SELECT * FROM USER_TAB_COMMENTS
```

Results | Explain | Describe | Saved SQL | History

TABLE_NAME	TABLE_TYPE	COMMENTS	ORIGIN_CON_ID
DEPARTMENTS	TABLE	-	3
DEPT	TABLE	This table contains information of the department	3
EMP	TABLE	This table contains information of all the employees	3
HTMLDB_PLAN_TABLE	TABLE	-	3

4 rows returned in 0.28 seconds Download

Q10. Drop the FIRST_NAME column from the EMP table. Confirm your modification by checking the description of the table.

SOLUTION:

```
-- Q10

ALTER TABLE EMP
DROP COLUMN FIRST_NAME
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table altered.
```

0.02 seconds

CHECKING THE DESCRIPTION OF THE TABLE:

```
DESCRIBE EMP
```

Results | Explain | Describe | Saved SQL | History

Object Type **TABLE** Object **EMP**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP	ID	NUMBER	-	7	0	-	✓	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	9	2	-	✓	-	-
	DEPT_ID	NUMBER	-	7	0	-	✓	-	-

Q11. In the EMP table, mark the DEPT_ID column in the EMP table as UNUSED. Confirm your modification by checking the description of the table.

SOLUTION:

```
--Q11

ALTER TABLE emp
SET UNUSED (dept_id);
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table altered.
```

0.06 seconds

CHECKING THE DESCRIPTION OF THE TABLE:

```
DESCRIBE EMP
```

Results | Explain | Describe | Saved SQL | History

Object Type **TABLE** Object **EMP**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP	ID	NUMBER	-	7	0	-	✓	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	9	2	-	✓	-	-

Q12. Drop all the UNUSED columns from the EMP table. Confirm your modification by checking the description of the table.

SOLUTION:

```
--Q12

ALTER TABLE EMP
DROP UNUSED COLUMNS;
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table altered.
```

0.07 seconds

CHECKING THE DESCRIPTION OF THE TABLE:

Results | Explain | Describe | Saved SQL | History

Object Type **TABLE** Object **EMP**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP	ID	NUMBER	-	7	0	-	✓	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	9	2	-	✓	-	-